

Prajwal Sahu

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LinkedIn | GitHub | Portfolio

Summary

Software Engineering student pursuing a B.E. in Computer Science at NIE Mysore (Sept 2023 -- Present), with hands-on experience in full-stack development, RESTful API design, scalable backend systems, and secure web application deployment. Proficient in Python, JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js, MongoDB, Redis, and CI/CD workflows. Strong foundation in data structures and algorithms, application security, and Agile development.

Education

The National Institute of Engineering

Bachelor of Engineering -- Computer Science | CGPA: 8.7

Mysore, India

Sept 2023 -- Present

Delhi Public School (DPS), Korba

High School Diploma -- Science (PCM) | 90%

Korba, Chhattisgarh

June 2021 -- May 2023

Experience

Tech Core Member -- OWASP NIE

September 2025 -- Present

Software Engineer / Open Web Application Security Project

Mysore, India

- Designed, developed, and maintained the **Linux CLI Handbook** using **Next.js 16**, **TypeScript**, and **shadcn/ui**; deployed on Vercel with **CI/CD** via GitHub Actions, achieving zero-downtime releases. [GitHub | Live]
- Implemented advanced search with **relevance scoring**, **fuzzy matching**, and **keyboard navigation**; applied **application security** best practices including input sanitization, secure headers, and XSS mitigation.
- Collaborated in an **Agile environment**; led the **Git-Gud** mentorship event on Git workflows for 50+ students; supported requirement gathering and version control best practices.

Projects

Viewly [GitHub | Live]

August 2025

- Built a **full-stack video streaming platform** with **React.js**, **Node.js**, and **Express.js**, supporting uploads, streaming, playlists, watch history, and creator analytics.
- Engineered **40+ RESTful APIs** across 11 controllers backed by **MongoDB** aggregation pipelines; integrated **Redis** caching and **Cloudinary** CDN, reducing API response time from **400 ms to 150 ms**.
- Secured the platform with **JWT** token rotation, bcrypt hashing, HTTP-only cookies, and XSS mitigation; managed global state across **22 pages** using **Redux Toolkit**.

Orbit Code [GitHub | Live]

January 2026

- Built a **multiplayer browser-based IDE** using **Next.js 16**, **TypeScript**, and **Monaco Editor** with real-time collaborative coding via an Operational Transformation (OT) engine -- **3 collaborators at 18ms latency**.
- Architected a **Docker-isolated sandboxed code execution** pipeline with **6 AI providers** (OpenAI, Gemini, Claude) via **Socket.io**; integrated AI chat sidebar and inline ghost-text completions.
- Built a **custom Git engine** with commit diffing, branch management, push/pull operations, and **GitHub API integration** for real remote repository sync directly from the browser IDE.
- Designed **Prisma ORM** models with **NextAuth v5** OAuth, **RBAC**, and CI/CD deployment on Vercel with COEP/COOP headers for SharedArrayBuffer support.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, shadcn/ui, HTML5, CSS3

Backend: Node.js, Express.js, Prisma ORM, Socket.io, RESTful APIs, JWT Authentication, bcrypt, WebSockets

Databases: MongoDB, SQLite, PostgreSQL, Redis

Cloud & DevOps: Docker, Vercel, Cloudinary, Linux, Git, GitHub Actions, CI/CD

Concepts: Data Structures and Algorithms, Object-Oriented Programming, Application Security, Agile, Unit Testing, Role-Based Access Control, Scalable Backend Systems